

# Metrics, scale, and governance

## IW Computer Vision Services

A companion briefing with the complete document flow

The project gain is a combination of scale, quality, operational efficiency, and controlled model improvement. We automate high-confidence work while preserving traceability and human oversight.

### Executive metric snapshot

| Dimension              | Documented project position                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Scale                  | Tens of thousands of documents per week across 20+ document classes, according to the project context.                   |
| Classification         | High-0.8 to 0.9 F1 is reported across document classes.                                                                  |
| Extraction             | Typically 0.7-0.9 field-level performance when text and location correctness are considered.                             |
| Newer GenAI flow       | 91%+ documented extraction accuracy across cited carrier scenarios: Aetna 91.7%, Cigna 93%+, Anthem 90%+.                |
| Dental table selection | F1, precision, and recall documented as 1.0 on 200 documents, 653 tables, and 4 categories.                              |
| Operational impact     | Portfolio reports under 10% manual review, under 5 seconds/document, and approximately \$0.03/document for the GPT flow. |

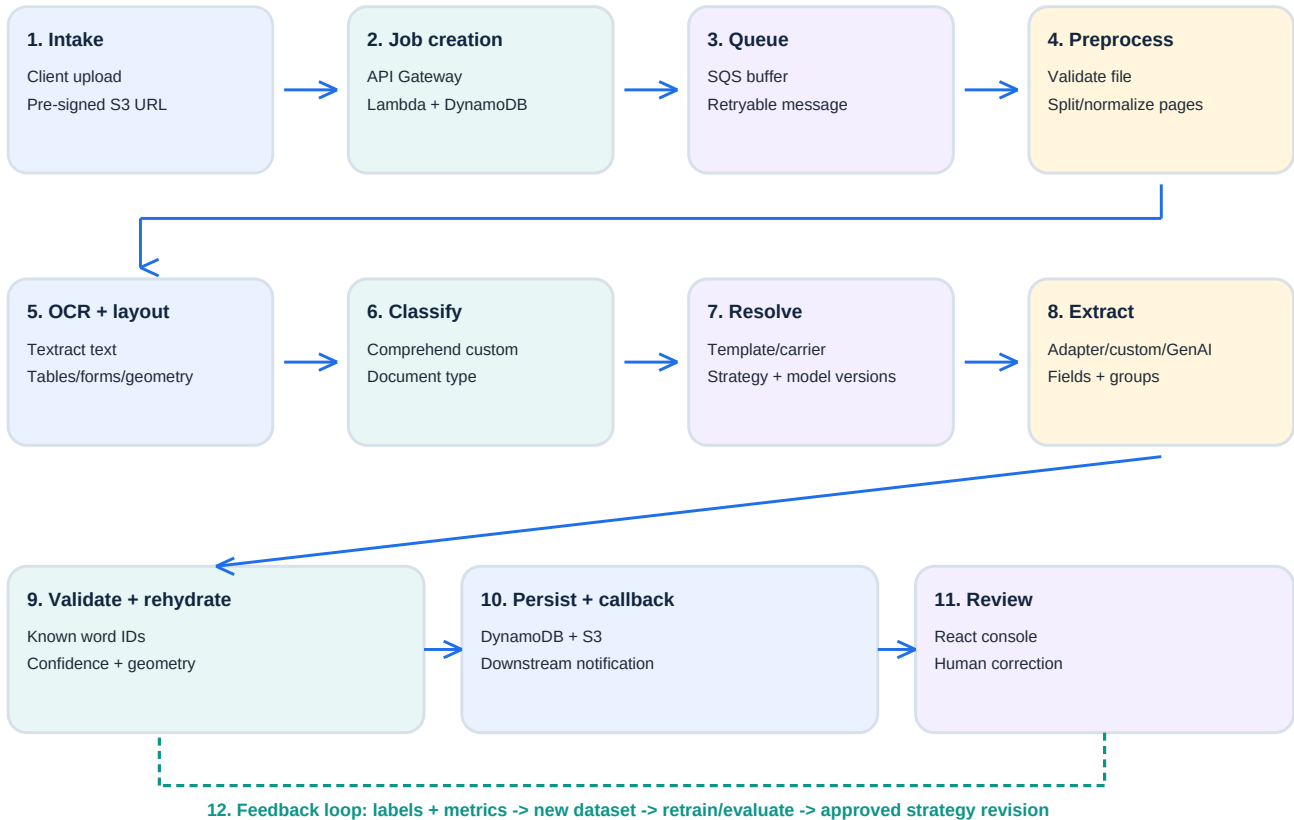
Important metric note: exact volume and accuracy claims should be presented with the carrier, cohort, fields, denominator, and evaluation date. The repository contains the platform and evaluation harnesses, while raw production volume counters are not included in this snapshot.

## 01

# Full flow: initial upload to final result

The complete inference and learning loop

This is the flow you can draw or explain in a presentation. The upper path is the document inference path. The lower path is validation, persistence, review, and the feedback loop that improves future revisions.



The important design idea is that the final result is not only a JSON payload. It includes job state, source document evidence, extracted fields/field groups, source word IDs and geometry, model/strategy identity, metrics, and a review path.

## 02

## What we gain from the metrics

Scale, quality, and business value

| Gain               | Meaning                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Throughput         | The asynchronous S3 -> SQS -> Lambda design lets the platform absorb large document volumes without requiring one synchronous request to remain open. |
| Automation         | Classification and extraction reduce the amount of manual locating, reading, and transcription work.                                                  |
| Quality            | Held-out evaluation, field-level metrics, spatial evidence, and human review make quality measurable rather than anecdotal.                           |
| Adaptability       | Carrier-specific behavior is isolated in strategy/profile configuration and versioned model components, reducing the cost of supporting new layouts.  |
| Traceability       | Word-ID grounding lets a reviewer trace a predicted value back to the exact source words and bounding boxes.                                          |
| Operational safety | Retries, failure states, explicit promotion, immutable revisions, and service-side validation reduce the risk of silently wrong production outputs.   |

## Recommended explanation of processing volume

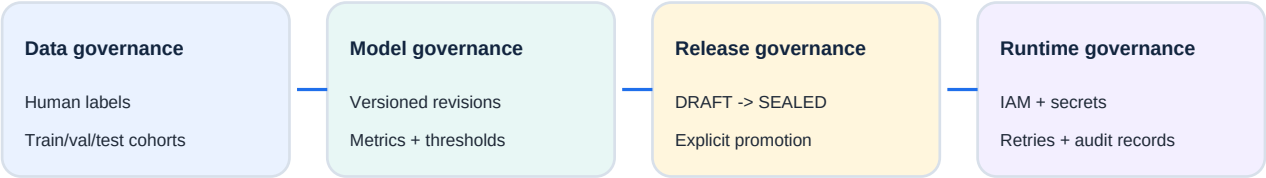
**We process tens of thousands of documents per week across more than 20 document classes. The platform handles the work asynchronously through queues and workers, so it can scale document processing independently from the user-facing API.**

Do not claim a specific documents-per-minute rate unless you have CloudWatch, SQS, or job-table evidence for that environment. Weekly volume is a business-scale statement; peak throughput and latency are infrastructure measurements.

03

# Model governance strategy

How quality is controlled before production



Governance is enforced in service code and persistence, not only in the UI.

| Control               | How it works                                                                                                                                   | Why it matters                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Dataset governance    | Human-labeled documents are selected into train, validation, and test cohorts. Test documents are kept out of training.                        | Prevents leakage and makes results comparable.                            |
| Model governance      | Models and extraction strategies have explicit IDs, versions, metrics, targets, and ownership.                                                 | Makes every output attributable to a specific revision.                   |
| Evaluation governance | Classification, extraction, threshold, field, document, and spatial metrics are materialized and reviewable.                                   | Shows not only whether the model is good, but where it fails.             |
| GenAI governance      | Prompts/profiles specify fields, output schema, examples, and known-word-ID contracts. Unknown IDs are rejected.                               | Constrains semantic flexibility and reduces hallucinated output.          |
| Promotion governance  | A strategy moves from DRAFT to SEALED only after required training/evaluation completes; only sealed revisions can be referenced by templates. | Prevents unreviewed experiments from becoming active production behavior. |
| Runtime governance    | IAM, Secrets Manager, server-side sessions, retries, terminal failure states, logging, and persistent job/result records.                      | Protects access and provides operational traceability.                    |

04

## Pitch paragraph

A clear answer to 'what did we gain?'

**We gained a scalable and governed document intelligence capability. At the reported operating scale, the platform processes tens of thousands of documents per week across 20+ document classes. It combines classification and extraction models with Textract OCR, carrier-aware strategies, and grounded GenAI. The documented results show high-0.8 to 0.9 classification F1, roughly 0.7-0.9 field-level extraction quality, and 90%+ extraction accuracy in the cited multi-carrier GenAI scenarios. The larger achievement is that these models are operated through a controlled lifecycle: human labels create cohorts, metrics determine quality, word IDs provide evidence, revisions are versioned, and only evaluated and sealed strategies can be promoted.**

### Short version

**We are not only gaining accuracy. We are gaining throughput, repeatability, traceability, faster onboarding of document variants, lower manual effort, and safer model governance.**

### Evidence reviewed

- Project READMEs and AWS/CDK architecture.
- Runtime inference flow and extraction strategy code.
- Composite extraction and strategy lifecycle design.
- GenAI extraction portfolio and carrier experiment documentation.
- React console components for review, metrics, and strategy authoring.

Metric guardrail: the quantitative results above are documented project figures. Before an external presentation, attach the evaluation cohort, sample size, metric definition, and source dashboard or report for each number.